Atomistic bounds on the magnetostrictive mechanism of magnetic-memory plasticity in Al–Mg–Si alloys with iron-bearing inclusions

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Abstract

How pre-exposing an Al–Mg–Si alloy with $Al_{13}Fe_4$ inclusions to a static magnetic field affects plastic deformation in creep is examined by molecular dynamics. The inclusion is elongated along the field by 0.194%, the strain corresponding to the 147 MPa interface stress estimated experimentally, and is held at that strain while the matrix relaxes. The resolved shear stress in the matrix reaches 15 MPa above the inclusion and falls to the resolution of the calculation within 80 Å. Under shear, a pre-existing dislocation pair is torn apart from 95–105 MPa, and no new dislocation forms at the interface up to 400 MPa in the unstrained cell. A two-scale estimate reproduces the measured 25% creep increase for the activation volumes reported for Al–Mg–Si, provided the inclusions strain by 0.194%; the magnetostriction measured for Fe–Al is twenty times smaller, and the persistence of the effect after the field is removed is not explained by an elastic stress.

Keywords: magnetoplasticity, magnetic memory, creep, molecular dynamics, Al–Mg–Si alloys, $Al_{13}Fe_4$, interface stress, dislocations, solute pinning

1. Introduction

Weak magnetic fields, whose energy per atom is negligible compared with the thermal energy kT, nevertheless change the plasticity of nonmagnetic crystals; the broad class of such observations is known as the magnetoplastic

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effect [1, 2]. Aluminum alloys with Fe-bearing inclusions occupy a special place in this class because, if such an inclusion is ferromagnetic, they offer a seemingly straightforward mechanical route from the field to plastic flow: the field magnetizes the inclusion; magnetostriction—the change of shape that a magnetized body undergoes—strains it; and the stress caused by the mismatch between the strained inclusion and the surrounding matrix is transferred to the matrix. Whether the inclusions in these alloys are in fact ferromagnetic is considered in Section 5.1. A series of experiments on a commercial Al–Mg–Si alloy with iron-bearing microinclusions has documented the effect in detail: a 30 min exposure to $B = 0.4$–0.7 T, applied *before* mechanical testing, increases the subsequent room-temperature creep strain—the slow deformation under a constant load—at 160 MPa by roughly a quarter, increases the heat release during deformation several-fold, and shifts the Taylor–Quinney coefficient, the fraction of the work of plastic deformation that is released as heat, from 0.07 to 0.25 [3, 4, 5, 6]. Because the effect remains after the field is removed, it has been described as "magnetic memory" in plasticity [4].

The interpretation advanced in these works rests on a single estimate. The stress at the inclusion–matrix interface produced by magnetostriction of the inclusion, obtained from the field strength through an empirical proportionality coefficient, is put at about 147 MPa for $B = 0.7$ T [5]. This exceeds the 120 MPa yield stress of the matrix, and the conclusion drawn is that the matrix around every inclusion is plastically deformed and its density of dislocations—the line defects of the crystal lattice whose motion produces plastic flow—is raised. This chain of reasoning has never been examined at the scale on which it is supposed to act. Whether a stress of this magnitude actually appears in the matrix, over what distance from the interface it persists, and whether it is capable of nucleating dislocations or of setting existing ones in motion are questions about ångströms and picoseconds—the native domain of classical molecular dynamics (MD), in which the atoms move according to Newton's equations under forces derived from an interatomic potential.

In the present work the magnetostrictive mechanism is subjected to a quantitative atomistic test at the interface between aluminum and the intermetallic compound $Al_{13}Fe_4$. A magnetic field has no direct representation in classical MD, so its action is represented mechanically: based on the earlier experimental work [5], the inclusion is taken to elongate along the field direction by $\varepsilon = 1.94\times10^{-3}$ (0.194%) at constant volume, the strain corresponding to the 147 MPa interface stress estimated there. A direct reproduction of the macroscopic effect in a simulation cell is not attempted; the reason is set out in Section 5. Instead, each link of the proposed chain is examined separately. The following quantities are computed: the stress field that the elongated inclusion creates in the matrix and its decay with distance from the interface; the applied shear stress at which a dislocation already present in the matrix starts to move, and that and whether new dislocations are nucleated at the interface up to the highest stress applied; and the resistance that Mg and Si solute atoms dissolved in the aluminum oppose to dislocation motion. The chain is then closed from the opposite end: a two-scale estimate gives the interface stress that the measured 25% creep enhancement *requires*, assuming thermally activated glide—dislocation motion in which obstacles are overcome with the help of thermal fluctuations, so that the glide rate depends exponentially on the stress—and the decay as r^{-3} of the elastic field outside a strained inclusion [7].

2. Computational methods

2.1. Interface cell

All calculations were performed with LAMMPS (22 Jul 2025 release, KOKKOS/CUDA build for graphics processors) [8]. Interatomic forces are given by the 2NN-MEAM potential (second-nearest-neighbour modified embedded-atom method) of Jelinek *et al.* for the Al–Si–Mg–Cu–Fe system [9], a single potential that describes every atom in the cell, including the atoms on either side of the interface, by one set of interaction rules.

The matrix is face-centred cubic (fcc) aluminium (a = 4.05 Å). Its orientation, $x \parallel [1\bar{1}0]$, $y \parallel [11\bar{2}]$, $z \parallel [111]$, is dictated by the slip system of fcc Al: dislocations in aluminium move on $\{111\}$ planes along $\langle 110 \rangle$ directions, and with this choice the (111) glide plane is parallel to the interface and the $[1\bar{1}0]$ glide direction lies along x. An edge dislocation (the edge of an extra half-plane of atoms inserted into the crystal) with Burgers vector $\frac{1}{2}[1\bar{1}0]$—the lattice translation that measures the displacement it carries, $|\mathbf{b}|$ = 2.86 Å—and line along y then spans the whole width of the cell, so that it represents an infinitely long straight dislocation, and glides in x parallel to the interface, which is the motion the mechanism under test would have to produce.

The inclusion is the monoclinic intermetallic $Al_{13}Fe_4$ in the structure of Ref. [10]. It is built as a flat layer 20 Å thick that carries on its upper surface a half-elliptical bulge (referred to below as the ridge) with semi-axes 35 Å along x and 20 Å along z, running the full length of the cell along y. The ridge is essential: a flat strained layer that spans the entire cell can relieve its normal stress simply by expanding upward, and it is the curved edges of the ridge that produce a non-uniform stress field in the matrix, as the curvature of the surface of a real particle would. The flat layer is left unstrained and the ridge alone carries the strain (Section 2.2).

In x and y the cell is periodic: an atom leaving through one face re-enters through the opposite face, so the cell represents an interface unbounded in its plane. The in-plane dimensions (L_x = 154.64 Å, L_y = 64.48 Å: 54 and 13 periods of the aluminium lattice, ten and eight cells of $Al_{13}Fe_4$) are chosen so that both lattices are periodic in the plane. The small residual mismatch (-0.22% along x, -0.26% along y) is taken up by the inclusion and is the same in the strained and in the control cell, so that it cancels to first order when the two are compared. Aluminium atoms lying within 2.2 Å of an inclusion atom (periodic images included) are removed; the smallest interatomic distance in the resulting structure is 2.22 Å. The aluminium layer above the interface plane is 129 Å thick (56 atomic (111) planes), the whole cell contains 91 428 atoms (72 532 in the matrix and 18 896 in the inclusion), and in z it is bounded by a free surface with 17 Å of vacuum above it and by a layer 6 Å thick at the bottom whose atoms are held fixed (Fig. 1).

The inclusion is bonded to the matrix. Atoms on both sides of the interface interact through the same potential, no constraint or artificial coupling is imposed at the interface, and the interface neither slips nor opens except through the atomic displacements the potential itself allows. The control structure is relaxed in three stages: energy minimisation by conjugate gradients (the atoms are moved to positions of minimum energy without thermal motion), 6 ps of molecular dynamics at 300 K followed by 12 ps of cooling to 0.5 K, which relax the long-wavelength modes of the slab that minimisation alone does not reach, and a final minimisation to a force of 0.05 eV/Å (two-norm over all atoms). The strained cell is built from this relaxed state (Section 2.2) and minimised to 0.02 eV/Å.

2.2. Representation of the field

Molecular dynamics is classical mechanics, and the magnetic field itself is not present in the calculation. Its action is represented mechanically, by the strain it is assumed to produce in the inclusion. Following the earlier experimental studies [3, 4, 5, 6], in which the field acts through magnetostriction,

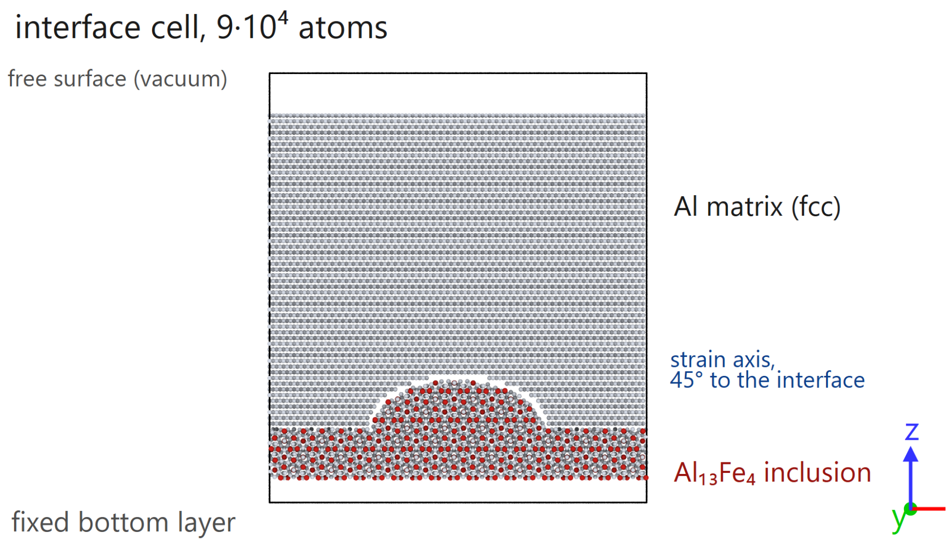

Figure 1: The interface cell, viewed along y (the x–z plane). Grey: the aluminium matrix, 129 Å of fcc aluminium with $x = [1\bar{1}0]$, $y = [11\bar{2}]$ and $z = [111]$, so that the interface is parallel to a (111) glide plane and x is a glide direction. Red: the $Al_{13}Fe_4$ inclusion, a flat layer with a half-elliptical ridge 70 Å wide and 20 Å high. The lowest 6 Å of the inclusion are held fixed; above the aluminium is vacuum. The cell is periodic along x and y. The imposed elongation of the inclusion is directed at 45° to the interface in the x–z plane.

i.e. the elongation of the inclusion along the direction of the field, the inclusion is taken to elongate along the field direction $\mathbf{u}$ by $\varepsilon = 1.94 \times 10^{-3}$ (0.194%), the elastic strain σ_m/E_{Al} (with $E_{\mathrm{Al}} = 75.7$ GPa) corresponding to the interface stress $\sigma_m = 147$ MPa estimated in those works [5], at constant volume: elongation ε along $\mathbf{u}$ and contraction $\varepsilon/2$ along the two perpendicular directions. Written as a tensor,

$$\varepsilon^*_{ij} = \tfrac{3}{2}\,\varepsilon\big(u_i u_j - \tfrac{1}{3}\delta_{ij}\big)\,, \qquad \operatorname{tr}\varepsilon^* = 0, \tag{1}$$

where δ_{ij} equals 1 for $i = j$ and 0 otherwise and the zero trace expresses the constancy of volume. In the elasticity of inclusions a strain of this kind, the one the inclusion would adopt if it were free of the matrix, is called an eigenstrain [7]; the term is used below in that sense. The value adopted is well above the magnetostriction measured in Fe–Al alloys [11, 12]; it is used because it is the amplitude the proposed mechanism itself presupposes, and the mechanism is tested at that amplitude.

The direction $\mathbf{u}$ is inclined at 45° to the interface normal, in the x–z plane. The reason is geometric. For $\mathbf{u}$ along the normal z the imposed strain produces no shear stress at all on the $(111)[1\bar{1}0]$ glide system: the Schmid factor (the geometric factor that converts a stress into the shear stress acting on a given slip system) is exactly zero, and no force whatever would act on a dislocation gliding parallel to the interface. At 45° the shear component of the imposed strain on the glide plane is at its maximum, $\varepsilon^*_{xz} = \frac{3}{4}\varepsilon = 1.46 \times 10^{-3}$. In a polycrystal the grains that respond to the field are those whose slip planes are favourably inclined to it, and the tilted cell represents that population. The tilt is the same in every cell.

The strain is applied by displacing every atom of the ridge relative to the centre of the ridge in proportion to its position, $\mathbf{r} \to \mathbf{r} + \varepsilon^*\!\cdot\mathbf{r}$, which stretches the ridge uniformly along $\mathbf{u}$ and compresses it across $\mathbf{u}$. The flat layer beneath the ridge is left unstrained. It spans the periodic cell, and a layer that is periodic in its plane cannot be sheared coherently: held at the shear of Eq. (1) its surface would be a sawtooth with a step of $\varepsilon^*_{xz}L_x = 0.22$ Å at the cell boundary, and the whole aluminium slab above it would be stretched along z by about ε^*_{xz}—a stress of order 150 MPa across the cell, which is what was observed when this was tried. The ridge is the particle whose field is measured; the layer only closes the cell. The interatomic potential is not modified. The stress field produced by the strained inclusion is obtained as the difference between a cell with the strained inclusion and a control cell without strain, both minimised from identical starting structures under identical conditions. Two variants are computed, differing in whether the inclusion is allowed to relax.

In the *maintained* case every inclusion atom—the ridge at its displaced position, the flat layer at its original one—is held by a stiff spring (stiffness 20 eV/Å^2, which keeps the atom within about 0.01 Å of that position) while the matrix relaxes around it, so that the ridge holds the strain of Eq. (1) throughout. The springs are attached in the same way in the corresponding control cell, so they cancel in the difference between the two. This cell gives the stress field of a particle that maintains its eigenstrain. Under the forces that arise at the interface the inclusion atoms depart from their strained positions by about 0.01 Å, and the ridge retains 0.97 ± 0.004 of the imposed strain (measured as described for the free case below).

In the *free* case no springs are attached and the inclusion is free to relax the imposed strain back out. It does so to a large extent: after minimisation the ridge, which produces the field measured in the matrix, retains 0.2–0.4 of the imposed strain (0.20 ± 0.11 and 0.44 ± 0.10 in two minimisations, both stopped before full convergence). The fraction is measured as the projection of the residual distortion of the Fe sublattice of the inclusion onto ε^*, obtained by a least-squares fit of the displacements between the strained and the control cell; the uncertainty is the standard error of that fit. The free cell shows what an inclusion does when nothing holds its strain; its stress field is the relaxed response, not the field of an inclusion that maintains the strain, and it is the maintained case that is compared with the dislocation thresholds below.

The *control* cell has no strain. It is the relaxed structure of Section 2.1 itself; the strained cell is built from it, minimised with the same tolerance and compared with it atom by atom, so that the residual lattice mismatch of Section 2.1, the arrangement of the interfacial layer and the long-wavelength strain of the slab are common to both and cancel in the difference. The interface cells supply the stress profile in the matrix and its decay length. The stresses at which dislocations begin to move are obtained separately, in the loaded cells of Section 2.3, and serve only as the thresholds against which that profile is compared.

2.3. Loaded cells and loading scheme

The stresses at which dislocations move are measured in two cells loaded in shear under dynamics at 300 K.

The first is the interface cell of Section 2.1 with a pre-existing pair of edge dislocations placed next to the ridge of the inclusion, 91 428 atoms in all (Fig. 2). The pair is a dislocation dipole: two parallel dislocations of opposite sign whose stress fields attract, so that in a uniform, stress-free crystal the pair is stable and stays where it is placed until the applied stress exceeds the level at which the two partners can pass each other. Both dislocations lie along y and glide on (111) planes parallel to the interface. The partners are offset by 23.4 Å along x and by ten (111) interplanar spacings, 23.4 Å, along z, so that the line joining them is inclined at 45° to the glide planes. Two edge dislocations of opposite sign on glide planes a distance h apart attract with a shear stress of up to $\mu b/[8\pi(1-\nu)h]$, which for $h = 23.4$ Å is 198 MPa (with $\mu = 26.5$ GPa, $\nu = 0.347$ and $b = 2.864$ Å); this is the applied stress at which they would pass each other in an unbounded crystal. The lower partner is placed 27 Å beyond the foot of the ridge and 29 Å above the interface plane, the upper one 23 Å higher and 23 Å closer to the ridge. The cell is loaded by a shear stress raised linearly from 0 to 400 MPa over 96 ps, after 5 ps at zero stress. The stress is applied as a uniform force along x on the atoms of the top atomic layers, with the reaction taken by the fixed bottom layer; the rate of rise is switched on smoothly over the first 16 ps so that the lowest shear vibration of the slab is not excited. The loading is far faster than any laboratory test (an effective strain rate of order 10^8 s^{-1}), so the thresholds obtained are upper estimates relative to slow loading. Three ramps were run: the unstrained cell with the inclusion free to relax, to 400 MPa over 96 ps; and the unstrained and the strained cell with the inclusion held by the springs of Section 2.2, at the same rate of rise, to 145 MPa over 40 ps—the pair that compares strained with unstrained.

The second cell has no inclusion. It is a slab of the alloy matrix with the same orientation, 92 800 atoms (91 315 Al, 928 Mg, 557 Si; 114.6 × 49.6 × 291 Å), in which 1.0 at.% Mg and 0.6 at.% Si are placed at random on lattice sites, and which contains one edge dipole whose partners lie 206 Å apart on parallel glide planes; their mutual attraction corresponds to a shear stress $\mu b/[8\pi(1-\nu)h] \approx 22$ MPa, small compared with the stresses applied. Before loading, the structure is energy-minimised and brought to 300 K. This cell is loaded in steps: constant shear stresses of 45, 55, 65 and 75 MPa, each held for 30 ps and each switched on smoothly over 4 ps, after 10 ps at zero stress. Steps are used rather than a ramp because a dislocation held by solute atoms moves by breaking away from them, a thermally activated event whose waiting time depends on the stress; holding the stress constant for a fixed time tests whether the dislocation breaks away at that level, and the stress is raised in steps to find the level at which it does. A pure-aluminium cell of the same geometry provides the solute-free reference.

In both cells the equations of motion are integrated with a time step of 1 fs, about a hundredth of the period of an atomic vibration. The temperature is held at 300 K by a Nosé–Hoover thermostat (damping time 0.1 ps), which keeps the average kinetic energy of the unconstrained atoms at the set value by a weak feedback on their velocities. The dislocations are inserted by the Volterra construction, in which the crystal is cut along a half-plane, the two faces are displaced relative to each other by one Burgers vector and rejoined, with the displacement field made compatible with the periodic cell. The dislocation content of every starting structure is verified with the dislocation extraction algorithm (DXA) [13], which locates dislocation lines in an atomic configuration and determines their Burgers vectors, as implemented in OVITO [14].

2.4. Stress evaluation

Local stresses are obtained from the per-atom virial stress, the atomic-scale counterpart of mechanical stress, computed for each atom from the forces between it and its neighbours and divided by the atomic volume $\Omega = 16.61$ Å^3: $\sigma_{ij} = -\langle W_{ij} \rangle / \Omega$. The stress of a single atom fluctuates thermally by several GPa, so no scalar measure is ever formed atom by atom. The six components of the tensor are first averaged over a slice 4 Å thick parallel to the interface (and, in the dynamic runs, over time frames), and only then are two scalar measures formed from the averaged tensor: the von Mises stress, a single positive number that measures the shear content of a stress state and enters the yield criterion of a ductile metal, and the resolved shear stress, the shear stress acting on a given slip plane in a given slip direction. $\mathrm{RSS}_{\max}$ denotes the largest resolved shear stress among the twelve $\{111\}\langle 110 \rangle$ slip systems of the matrix, i.e. the shear stress on the most favourably oriented system.

Stress profiles $\sigma(r)$ are built as functions of the distance r from the interface plane in slices 4 Å thick, using only matrix atoms that lie above the crest of the ridge, so that r measures the distance from the inclusion surface. The field of the strained inclusion is the difference between the strained and the control cell, slice by slice. On energy-minimised structures there is no thermal motion, and the noise level of that difference is set by the residual slice-to-slice scatter far from the interface: over $r \geq 60$ Å the mean of

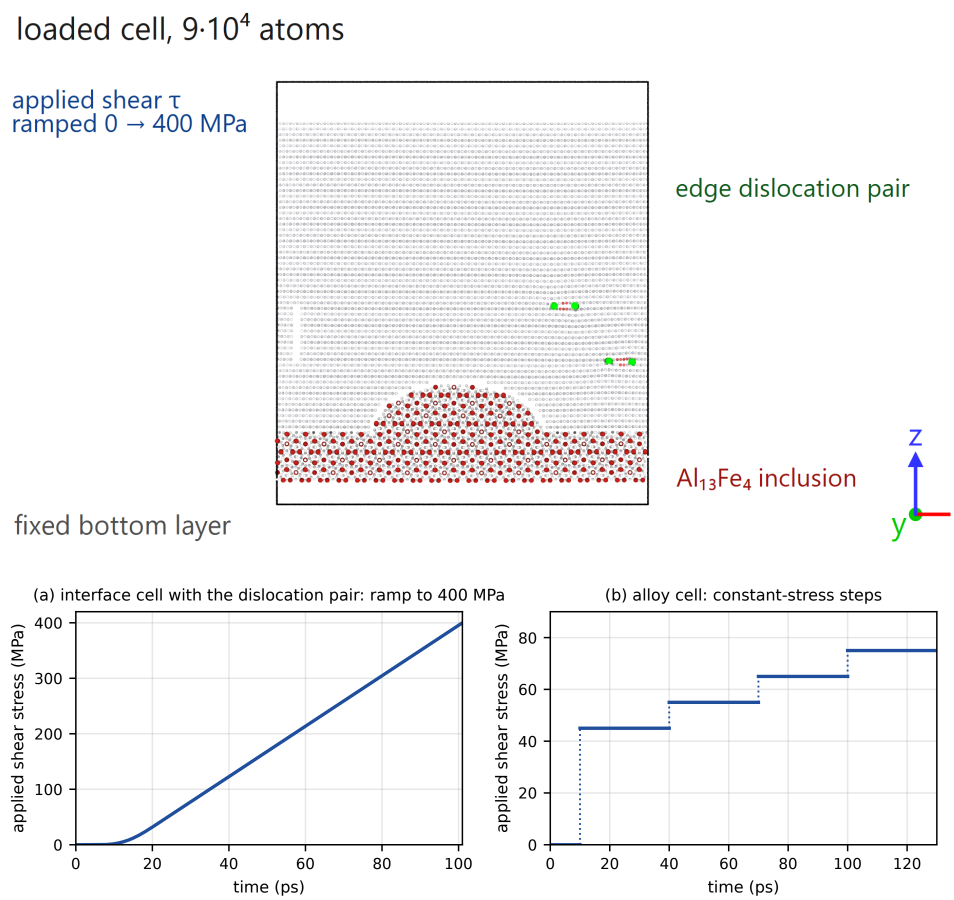

Figure 2: The loaded interface cell (top) and the two loading programmes (bottom). Top: the cell of Fig. 1 with a pair of edge dislocations of opposite sign next to the ridge (green: the dislocation lines found by dislocation analysis; atoms of the perfect lattice are faded), both on (111) glide planes parallel to the interface. A shear stress τ along x is applied through a uniform force on the top atomic layers and is taken up by the fixed bottom layer. Bottom: the stress programmes—for this cell a linear rise from 0 to 400 MPa over 96 ps after 5 ps at zero stress (the held cells were ramped at the same rate to 145 MPa) (a); for the alloy cell without inclusion (Section 3.3) steps of 45, 55, 65 and 75 MPa, 30 ps each, after 10 ps without load (b).

ΔRSS_{max} is 2.5±0.5 MPa (mean and standard deviation over thirteen slices), the resolution of the calculation, below which no field is resolved.

3. Results

3.1. The interface stress field and its decay

Figure 3 shows how the stress in the aluminium matrix varies with the distance r from the interface plane in the two interface cells of Section 2.1: the control cell, in which the ridge is unstrained, and the cell in which the ridge has been elongated by $\varepsilon = 1.94 \times 10^{-3}$ along the field direction and held there (Section 2.2). The strained cell is built from the relaxed control by displacing the ridge atoms and is then minimised, so that the two cells differ in nothing but the strain of the ridge. The matrix is cut into slices 4 Å thick parallel to the interface, the six components of the stress tensor are averaged over the aluminium atoms of each slice, and only then are invariants formed from the averaged tensor (Section 2.4). Two averages are used. The first spans the whole cell in x and y, i.e. the region above the ridge and the region beside it; the second is restricted to a window 20 Å wide centred on the ridge axis, which is the stress that the matrix sees directly above the inclusion. The ridge—the half-elliptic bump of the inclusion, with semi-axes 35 Å along x and 20 Å along z (Fig. 1)—has its crest at $r = 20$ Å, so that above the crest the distance to the nearest inclusion surface is $d = r - 20$ Å.

The upper panel gives the von Mises stress (Section 2.4) in each cell. Two features of the control curve need to be stated plainly, because they are present without any imposed strain. First, in the slices just above the ridge crest the stress is 100–190 MPa, and at the level of the ridge flanks it reaches 0.8–2.2 GPa. This is the coherency stress of the interface. Aluminium and $Al_{13}Fe_4$ are different crystals with different interatomic spacings; across the interface their atoms interact through the same interatomic potential as inside either crystal, with no constraint of any kind (Section 2.1), so the interface is bonded and cannot slip, and the atoms on both sides are pushed away from the positions their own lattice would assign them. It falls below 10 MPa beyond $r \approx 65$ Å and below 2 MPa beyond about 95 Å; the alternation between neighbouring slices closer to the crest reflects the bending of the atomic planes over the crest, where a 4 Å slice holds one or two planes whose stresses differ. This stress exists whether or not the ridge is strained: the two curves of the upper panel coincide, the difference tensor between them having a von Mises stress of at most 9 MPa above the crest.

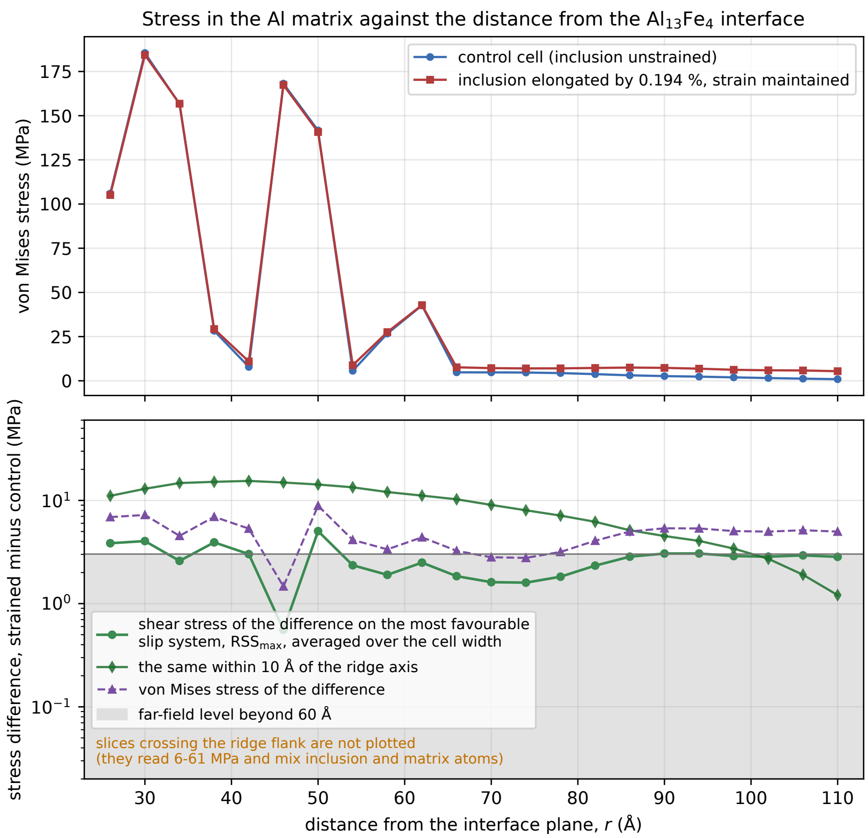

Figure 3: Stress in the aluminium matrix as a function of the distance r from the interface plane in the two interface cells (91 428 atoms): the control cell with the unstrained ridge, and the cell in which the ridge is elongated by 0.194% along the field direction and held at that strain. The matrix is cut into slices 4 Å thick parallel to the interface; the six stress components are averaged over the aluminium atoms of each slice before any invariant is formed. Top: the von Mises stress in each cell; the 100–190 MPa just above the ridge crest, present in both cells, is the coherency stress of the bonded interface between two crystals of different spacing, and the curves coincide. Bottom: the difference $\Delta\sigma_{ij} = \sigma_{ij}(\varepsilon) - \sigma_{ij}(0)$ as the shear stress resolved onto the most favourable of the twelve slip systems of fcc Al, $\mathrm{RSS}_{\max}$, averaged over the whole width of the cell (circles) and within 10 Å of the ridge axis (diamonds), together with the von Mises stress of the difference tensor. The shaded region marks stresses below 3 MPa: the width-averaged level beyond 60 Å with its scatter, 2.5 ± 0.5 MPa, the resolution of the calculation for long-wavelength strains of the slab. Slices at the level of the ridge flanks, closer to the interface plane than the crest, are omitted: there r is not a distance to the inclusion surface and the slices contain inclusion atoms. The profile ends 20 Å below the free surface of the slab.

That is why the effect of the imposed strain is measured throughout as the difference between the strained and the control cell, slice by slice. Second, the profile ends 20 Å below the free surface of the aluminium slab, which lies at $r = 129$ Å; in the slices nearer the surface the per-atom stress is dominated by the surface layers themselves.

The lower panel shows the difference, $\Delta\sigma_{ij}(r) = \sigma_{ij}(r;\varepsilon) - \sigma_{ij}(r;0)$, which is the quantity that the mechanism of Ref. [5] requires to exceed the yield stress of the matrix. Every threshold for dislocation motion in Section 3.2 is a resolved shear stress, so the difference is expressed in the same way. $\mathrm{RSS}_{\max}$ (Section 2.4) is computed from the difference tensor, not as a difference of von Mises values, which would not be an invariant of $\Delta\sigma_{ij}$; the von Mises stress of the difference tensor is plotted alongside it. In every slice above the crest but one the most favourable system is $(111)[1\bar{1}0]$: the glide plane parallel to the interface and the direction x, along which the dislocations of Section 3.2 glide, for which the resolved shear is simply the component $\Delta\sigma_{xz}$. The exception is the slice at $r = 46$ Å, where the two contributions nearly cancel and the largest of the twelve values, 0.6 MPa, falls on $(111)[01\bar{1}]$.

Above the ridge axis the resolved shear stress is 11–15 MPa from the crest out to $d \approx 30$ Å, with its maximum of 15 MPa at $d = 22$ Å; it then decays, to 10 MPa at $d = 46$ Å, 5 MPa at $d \approx 65$ Å and 2 MPa at $d \approx 85$ Å. Averaged over the whole width of the cell the same difference is much smaller, up to 5 MPa (maximum 5.0 MPa at $r = 50$ Å, one slice at 0.6 MPa), because the ridge occupies less than half of the width and the shear beside the ridge has the opposite sign to the shear above it. Beyond 60 Å the width-averaged $\mathrm{RSS}_{\max}$ settles at 2.5 ± 0.5 MPa (mean and standard deviation over thirteen slices). This level is the resolution of the calculation rather than a field of the ridge: a uniform shear of 2 MPa across the slab exerts a force of 10^{-4} eV/Å on each atom, below what the minimisation resolves, and it is with this level that the on-axis field merges at $d \approx 80$ Å. The stress that the strained ridge sends into the matrix is therefore local: 15 MPa directly above it within one ridge height, and a third of that on average across a region twice the width of the ridge.

The field is compared in Fig. 4 with the exact solution of Eshelby [7] for a spherical inclusion of the same radius, 35 Å, held at the same strain in an unbounded aluminium matrix: 41 MPa inside the sphere (47 MPa just outside its surface), 11 MPa at $d = 17$ Å, 5 MPa at $d = 35$ Å, falling as the inverse cube of the distance from the centre. The atomistic ridge produces a smaller stress at its surface and a flatter profile. The smaller value is a matter of shape: a half-elliptical bump on a flat layer radiates less into the matrix above it than a compact particle does (a cylinder holding the same strain carries almost the same interior stress as the sphere, 39 against 41 MPa); the flatter decay is that of a body infinite along y, whose field falls off more slowly than the inverse cube of a compact particle. On the ridge axis the two agree within a factor of two between $d = 10$ and 26 Å. They agree in magnitude only there: closer in the sphere is the larger, and beyond $d \approx 14$ Å the ridge is, until at $d = 50$ Å it exceeds the sphere threefold. The width-averaged curve is not comparable with either in this way, because slices in which the two contributions nearly cancel alternate with slices in which they do not. A two-dimensional elastic calculation for the ridge alone, in an unbounded matrix of the same stiffness, gives 20 MPa at the ridge surface falling to 5 MPa within 10 Å (Section 5.4). The retained fraction of the imposed strain, measured by fitting the residual distortion of the Fe sublattice of the ridge to ε^* (Section 2.2), is 0.97 ± 0.004 in the held cell. When the springs are removed and the inclusion is free to relax, the ridge retains only 0.2–0.4 of the strain (0.20 ± 0.11 and 0.44 ± 0.10 in two minimisations stopped before full convergence) and the shear stress on the ridge axis falls below 5 MPa; the two free minimisations then also differ across the whole slab, by 10 ± 3 MPa beyond 60 Å against 2.5 ± 0.5 MPa for the held pair, because an unconstrained inclusion relaxes along the normal and shifts the stress of the slab as a whole. The free cell therefore gives the retained strain but no usable stress field, and the field of Fig. 3 exists only as long as something holds the inclusion at its strain.

Slices at the level of the ridge flanks, closer to the interface plane than the crest, are omitted from the figure. There r measures height above the interface plane and not the distance to the inclusion surface, which differs for every atom in the slice; the coherency stress is 0.8–2.2 GPa, and the difference, 6–61 MPa from one 4 Å slice to the next, mixes the response of the matrix with the displaced inclusion atoms themselves. These slices characterise the interface, not the field it sends into the matrix, and they are not carried forward.

Two qualifications apply. The local stress is deliberately not compared with the macroscopic yield stress $\sigma_Y \approx 120$ MPa: a stress averaged over a 4 Å slice is not a quantity that can be set against a yield stress measured on a specimen, and the stresses that matter for dislocations are measured directly in Section 3.2. Whether the imposed strain plastifies the matrix is therefore decided by comparing the difference profile with those measured onsets, not by a local yield criterion; the interface cells are built without dislocations, and dislocation analysis of the strained cell after minimisation finds none. Second, the range of the field—some 30 Å at its full strength and 80 Å in all—is set by the size of the ridge. This accords with a general property of the Eshelby solution: the stress inside an inclusion does not depend on its size, whereas the reach of its exterior field scales with the size [7], so that a micron-sized inclusion held at the same strain would carry the same stress over a fraction of a micron—a point taken up in Section 4.

3.2. Thresholds for dislocation activity at the interface

The stresses at which dislocations respond were measured in the loaded cell of Section 2.3, which contains the same ridge together with a pre-existing pair of edge dislocations of opposite sign (a dislocation dipole) on glide planes parallel to the interface, under an applied shear stress rising linearly from 0 to 400 MPa in 96 ps (Fig. 2). The position of each dislocation was followed frame by frame, every 2 ps, by dislocation analysis. The onset of motion is defined as the stress at which the position departs from its thermal oscillation by more than six times the oscillation amplitude (and by at least 8 Å) and does not return, either because the line keeps moving or because it ceases to exist. One frame of the ramp corresponds to 9 MPa, which is the resolution of every onset quoted.

In the control cell, with the ridge unstrained and the inclusion free to relax during loading, the two partners behave differently from the start. The upper partner sits 32 Å above the crest of the ridge, in the coherency stress of the interface (Section 3.1); it is not at rest at the position where it was placed and glides 38 Å away from the ridge within the first 6 ps, at zero applied stress, after which it hovers within a few ångströms of its new position. The lower partner, which by the start of the ramp sits 32 Å beyond the foot of the ridge, settles by 13 Å in the same interval and then holds. It starts to move at an applied shear of 95–105 MPa, crosses the periodic boundary of the cell and within the next 2 ps is no longer recognised as a dislocation line: it has reacted with the interface. The upper partner, now unopposed, departs at 115–125 MPa and is gone in the same way by 180 MPa. No new dislocation appears at the $Al/Al_{13}Fe_4$ interface up to the end of the ramp at 400 MPa: heterogeneous nucleation at the interface—the creation of a dislocation at the interface rather than in the perfect lattice—needs more than three times the macroscopic yield stress of the alloy (120 MPa) in this cell. The two do not contradict each other: the yield stress of a specimen reflects the motion of the dislocations it already contains. The only new segments—transiently at 232–241 MPa and again from 377 MPa on—are partial dislocations 10–25 Å long at the periodic boundary of the cell, at the seam left by the insertion of the pair; none is at the interface, and they are a construction artefact.

With the inclusion held by springs, as in the interface cells, the picture changes in one respect and not in the other. The lower partner is then not held even at zero applied stress: in the control cell and in the strained cell alike it is gone by 10 ps, through the periodic boundary, because the coherency field of a rigidly held, unrelaxed inclusion is stronger than that of one free to relax. The upper partner moves 31–33 Å over the ridge in the first 6 ps in both cells, coming to rest 7–10 Å from its axis, and departs at 105–115 MPa in the control and 115–125 MPa in the strained cell, one frame of the ramp apart; these two ramps were carried to 145 MPa, and by 140 MPa the strained cell contains no dislocation while the control retains only fragments 13–26 Å long at the periodic seam. A short segment at that seam ($x = 7$–10 Å, $z = 57$–70 Å) is present in both held cells from 12–14 ps on, the same construction artefact as in the free ramp. The strain of the ridge thus changes the fate of the pair by at most one frame, 9 MPa. This is the expected result: the field of the strained ridge at the initial positions of the partners is 5–10 MPa, and on the ridge axis at the height where the upper partner comes to rest, $d \approx 32$ Å, it is 13–14 MPa (Section 3.1)—both of the order of one frame of the ramp, so a shift of one frame is neither resolved nor excluded.

The first picoseconds of every ramp, at zero applied stress, answer the question whether the field of the strained ridge alone moves a dislocation placed next to it: the same partner moves in the same way with the ridge strained and unstrained, driven by the coherency field of the interface, not by the strain of the ridge. All onsets are specific to this cell, to the single loading rate of order 10^8 s^{-1} (relative to quasistatic loading they are upper estimates), and to the interatomic potential; they are not material constants. The stress at which the lower partner breaks away, 95–105 MPa, is of the order of the attraction between the two partners at their separation ($\mu b/[8\pi(1-\nu)h] =$ 198 MPa for $h = 23.4$ Å, Section 2.3), reduced by the coherency field that has already pulled them apart; it lies below the macroscopic yield stress of the alloy, 120 MPa, because a pre-existing dislocation in a perfect matrix has nothing but its partner to hold it.

Set against these onsets on the common footing of resolved shear stress, the stress that the held ridge produces just above its crest, 15 MPa (Section 3.1), is a factor of 6–7 below the stress at which the lower partner starts to move and more than 25 below the highest stress reached without any dislocation forming at the interface (400 MPa), and it falls to the far-field level within 80 Å of the inclusion. The strained ridge therefore neither moves the pair nor creates dislocations by itself, and its bias on the pair is below the resolution of the ramp, one frame or 9 MPa, which is the order of the 5–14 MPa it produces where the pair sits. Figure 4 places the stress profiles and the onsets on one plot.

3.3. Dislocation mobility and solute pinning in the alloy

The resistance that dissolved Mg and Si atoms oppose to dislocation motion was measured in the alloy cell of Section 2.3: no inclusion, 1.0 at.% Mg and 0.6 at.% Si substituted at random for aluminium, and one pair of edge dislocations of opposite sign on glide planes 206 Å apart, loaded by a staircase of constant applied shear stresses of 45, 55, 65 and 75 MPa, 30 ps each. Figure 5 shows the position of the dislocation whose motion is followed (the lower member of the pair) and of its partner as a function of time. The dislocation does not glide. Over the 120 ps of loading it advances 2.0 Å—less than one Burgers vector $b = 2.86$ Å—against a thermal oscillation of 0.6 Å; the net displacements within the four rungs, 0.6, 1.6, -1.5 and 0.2 Å at 45, 55, 65 and 75 MPa, are comparable to that oscillation and do not grow with stress. The line oscillates about one position and remains held by the solute atoms around it at every stress level. A velocity can of course be fitted to each rung, but at these displacement-to-oscillation ratios (all below 3) it would measure the oscillation and not a drift, and none is quoted. What the trajectory does establish is a lower bound: the sampled random configuration of Mg and Si atoms holds the dislocation against a driving stress of at least 75 MPa. This exceeds the 15 MPa that the held ridge produces at its surface (Section 3.1) by a factor of 5, and the 41 MPa of the analytical sphere by a factor of 1.8. The two dislocations of the pair exert a mutual stress of 22 MPa on each other, whose sign relative to the applied shear is not resolved here, so the conservative reading of the bound is 75 ± 22 MPa, with a lower edge of 53 MPa, which still exceeds both amplitudes. The bound also lies above the Labusch estimate $\tau_c \approx 38$ MPa for this solute content [15, 16]—the mean-field estimate of the stress needed to drag a dislocation through a random distribution of solute atoms—which is consistent with a locally stronger-than-average arrangement of solutes in this single realization.

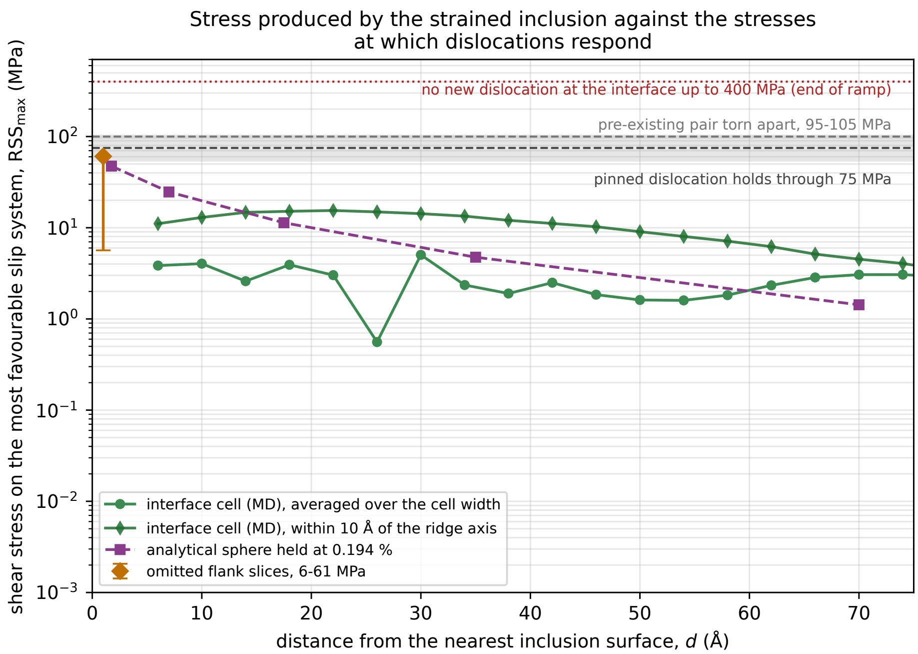

Figure 4: The stress produced by the strained inclusion against the stresses at which dislocations respond: the shear stress resolved onto the most favourable slip system, $\mathrm{RSS}_{\mathrm{max}}$, on a logarithmic scale, against the distance d from the nearest inclusion surface. Green circles: the interface cell (91 428 atoms) with the ridge held at 0.194%, averaged over the width of the cell; green diamonds: the same within 10 Å of the ridge axis; for the ridge $d = r - 20$ Å (the crest height). Purple squares: the exact solution of Eshelby [7] for a spherical inclusion of radius $a = 35$ Å held at the same elongation in an infinite elastic aluminium matrix ($\mu = 26.5$ GPa, $\nu = 0.347$, the volume-conserving elongation of Eq. (1) with its axis at 45° to z, the same maximum over the twelve slip systems); for the sphere $d = r - a$. Horizontal lines: the end of the ramp, 400 MPa, up to which no new dislocation formed at the interface (dotted), and the stress at which the pre-existing dislocation pair starts to move (95–105 MPa), both from Section 3.2; the shaded band, 75 ± 22 MPa, is the stress withstood by the pinned dislocation of Section 3.3 (the applied 75 MPa, plus or minus the 22 MPa mutual stress of the pair, whose sign is not resolved). The orange diamond marks the range spanned by the omitted slices on the ridge flanks; it is not a measurement of the field.

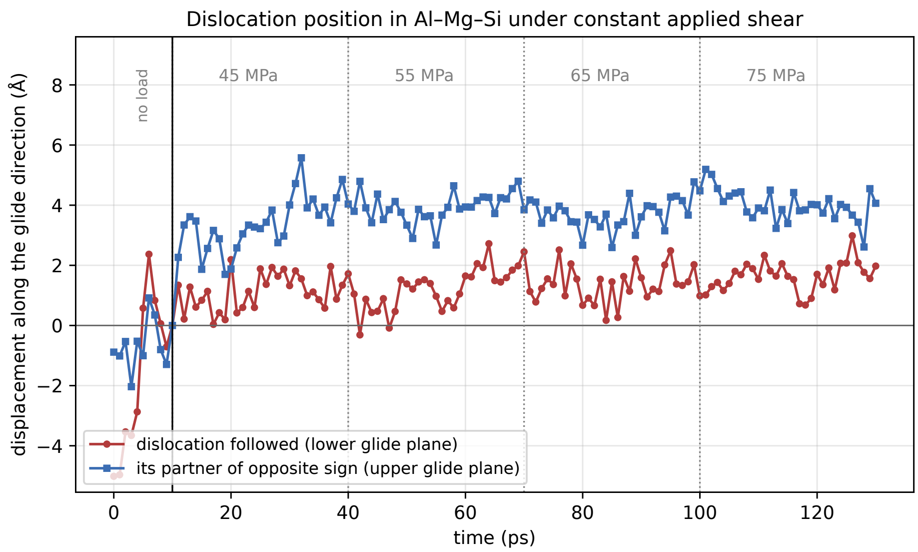

Figure 5: Position of the dislocation along its glide direction as a function of time in the Al–Mg–Si cell under the staircase of constant applied shear stresses (dotted lines: rung boundaries; the applied stress is written above each rung; the displacement is measured from the position at the moment the load is switched on, $t = 10$ ps, after 10 ps without load). Red: the dislocation whose motion is followed (the lower member of the pair); blue: its partner of opposite sign on the upper glide plane. The dislocation oscillates about one position and does not move: there is no plastic displacement at any of the four stress levels. Over the whole loading history it advances 2.0 Å, less than one Burgers vector $b = 2.86$ Å, against a thermal oscillation of 0.6 Å. The partner takes up 4.1 Å $(1.4\,b)$ when the load is switched on and then stops: over the remaining three rungs its position changes by less than its thermal oscillation of 0.5 Å. That single initial step is not sustained glide.

In the solute-free reference cell of the same geometry the dislocation moves freely: its position wanders by up to about 20 Å within individual stress steps, and no trapping event occurs in the 130 ps trajectory. Within this model, pinning by solute atoms is therefore the only obstacle to dislocation motion represented in these cells. A quantitative law of dislocation drag in pure aluminium could not be extracted from the same geometry: without obstacles one member of the pair reaches the free surface and is absorbed within the first stress step, and the excursions of the remaining line are bounded by the residual field of the periodic dislocation construction; the pinning observation in the alloy is therefore benchmarked against the Labusch estimate rather than against a measured drag coefficient. Because no depinning event occurs, the activation volume V^* (the volume swept by a dislocation segment as it crosses an obstacle, which sets how strongly the glide rate responds to stress) cannot be extracted from these runs. The value $V^* = 70\,b^3$ reported by Soula et al. for Al–Mg–Si processed by equal-channel angular pressing [17] is used as the central case; because that material state differs from the present one, the range $V^* = 19$–$142\,b^3$ [18] is used only as a sensitivity range. The trajectory serves solely to bound the obstacle strength from below. No Mg/Si clusters were constructed or identified. A second random configuration, run only through the 45, 55 and 65 MPa rungs, was likewise pinned at every rung it reached; the 75 MPa bound rests on the one realization that completed the staircase.

4. From local resolved shear to macroscopic creep

The results of Section 3 bound what a strained inclusion can do locally. To connect them with the macroscopic observation the question is turned around: how large a stress around the inclusions would be needed to raise the creep rate by the measured 25%? The estimate that answers it has three ingredients.

The first is the shape of the stress field. Outside a compact particle of radius a that is bonded to the matrix and strained, the elastic solution of Eshelby [7] gives a stress that falls off with distance r from the particle centre as $(a/r)^3$. The resolved shear stress—the part of the stress that pushes a dislocation (Section 2.4), and hence the only part that pushes a dislocation—is written in this scalar form as $\tau(r) = \tau_m(a/r)^3$, with τ_m its value at the particle surface. In a dilute dispersion with inclusion volume fraction f, the fraction of matrix volume in which $|\tau|$ exceeds a given level s is $F(> s) =$

$f\,(\tau_m/s-1)$ for $0<s<\tau_m$ (the volume of the particle itself is subtracted), and the distribution of stress levels over the matrix volume is given by the density $p(s) = -\mathrm{d}F/\mathrm{d}s = f\,\tau_m/s^2$.

The second is the response of dislocation glide to stress. At room temperature dislocations pass obstacles by thermally activated jumps, and a local stress τ changes the jump rate by the factor $\exp(\pm V^*\tau/kT)$. Here V^* is the activation volume of Section 3.3 [18]. Because the stress around a particle takes both signs, the two directions are weighted equally and the rate is proportional to $\cosh(V^*\tau/kT)$.

The third is the averaging. The creep rate of the alloy is the rate averaged over the matrix volume, so $\cosh(V^*\tau/kT)-1$ is integrated over the density $p(s)$. The relative enhancement of the creep rate is

$$\frac{\langle\dot{\varepsilon}\rangle}{\dot{\varepsilon}_0} - 1 = f\,x_0 \int_0^{x_0} \frac{\cosh u - 1}{u^2}\,\mathrm{d}u = f\left[x_0\,\mathrm{Shi}(x_0) - (\cosh x_0 - 1)\right], \qquad (2)$$

with $x_0 = V^*\tau_m/kT$, where $\mathrm{Shi}(x) = \int_0^x \sinh u\,\mathrm{d}u/u$ is the hyperbolic sine integral. The integral is taken down to $s=0$, i.e. it assumes that the stressed shells of neighbouring particles do not overlap; cutting it off where they would (at $u_{\min} = f x_0$, where the shell of one particle reaches its neighbours) lowers the predicted enhancement from 0.2500 to 0.2499 and raises the required τ_m by about 0.01% at every V^* in Table 1, so the tabulated values stand. Equation (2) is a model estimate of sensitivity, not an extraction of the interface stress from the experiment: it describes the field by a single amplitude τ_m, weights positive and negative stresses equally, and takes the spatial statistics of the $(a/r)^3$ field. It matters that τ_m is the amplitude of the resolved shear at the particle surface, not the full interface stress, of which only a part acts in any one slip plane.

The inputs are the following. The temperature is 300 K. The inclusion content of the alloy is 0.35 wt%, corresponding to a volume fraction $f = 0.00246$, within the 0.1–0.3% reported in Ref. [5]; Table 1 is evaluated at this value. The activation volume is taken in the range $V^* = 19$–$142\,b^3$, which spans the values for aluminium alloys [18], with $V^* = 70\,b^3$ measured for Al–Mg–Si [17] as the central case; no value of V^* was extracted from the present simulations (Section 3.3), so the range is a sensitivity range within the model, not a measured input.

The output is Table 1. Its second column is obtained by setting the left side of Eq. (2) to 0.25 and solving for τ_m: 8.4–62.7 MPa across the range of V^*, the larger values at the smaller activation volumes. Although the enhancement is proportional to f at fixed τ_m, the required τ_m is obtained by inverting a nonlinear function and therefore does not scale as $1/f$: direct recalculation gives 9.8–73.3 MPa at $f = 0.001$ and 8.1–60.3 MPa at $f = 0.003$.

Three comparisons follow. If the 147 MPa of Ref. [5] were the resolved shear stress acting on the dislocations, Eq. (2) would predict enhancements from 6.8×10^2 to 2.7×10^{46} (third column), at least 2.7×10^3 times the observed 0.25; 147 MPa acting as a shear stress is incompatible with the observed enhancement. The stress that an inclusion strained by 0.194% actually produces is smaller. For a bonded sphere held at that strain the exact solution [7] gives 41 MPa of resolved shear inside the particle and just outside it (Section 3.1); with this amplitude Eq. (2) gives 0.045 at $V^* = 19\,b^3$, 0.31 at $30\,b^3$ and values far above 0.25 at larger V^* (fourth column). The ridge of the interface cell, held at the same strain, produces 15 MPa of resolved shear just above its crest (Section 3.1); with this amplitude the estimate gives 0.042 at $V^* = 50\,b^3$ and 0.16 at $70\,b^3$, the activation volume measured for Al–Mg–Si [17] (last column). At a strain of 0.194%, therefore, the stress around the inclusions is of the order the estimate requires: the observed 0.25 is reproduced for V^* between about 30 and $75\,b^3$, depending on which of the two amplitudes is taken.

This is a consistency, not a confirmation, for three reasons. The estimate is exponentially sensitive to the product $V^*\tau_m$: doubling V^* at fixed amplitude moves the prediction by two to four orders of magnitude, so the agreement fixes little beyond the order of magnitude. The strain of 0.194% is an input taken from the earlier estimate of the interface stress [5], not a measured property of the inclusions; the magnetostriction measured for bulk iron–aluminium alloys does not exceed 10^{-4} [11, 12], twenty times less, and since the stress is proportional to the strain an inclusion strained by 10^{-4} would produce at most 2 MPa at its surface, for which Eq. (2) gives an enhancement below 0.4% at every V^* in the range. And the estimate describes a field that exists while the inclusion is strained; it says nothing about the tens of minutes for which the effect persists after the field is removed (Section 5.3). Whether the inclusions of this alloy strain by 0.194% in a field of 0.7 T is therefore the question on which the elastic mechanism stands or falls, and it is an experimental one.

Table 1: The two-scale estimate, Eq. (2), at $T = 300$ K, $f = 0.00246$ and $b = 2.8638$ Å. Second column: the resolved shear stress τ_m at the particle surface that reproduces the measured +25% creep enhancement. The remaining columns give the enhancement $\langle \dot{\varepsilon} \rangle / \dot{\varepsilon}_0 - 1$ that Eq. (2) predicts if τ_m were 147 MPa (the earlier estimate), 41 MPa (the interior value of the analytical sphere held at 0.194%, which is also the amplitude of its inverse-cube far field; 41.45 unrounded) or 15 MPa (the maximum on the ridge axis in the interface cell, 15.45 unrounded, Section 3.1). The observed value is 0.25.

V^*	τ_m for +25% (MPa)	at 147 MPa	at 41 MPa	at 15 MPa
$19\,b^3$	62.7	6.8×10^2	0.045	0.004
$30\,b^3$	39.7	3.9×10^6	0.31	0.010
$50\,b^3$	23.8	3.9×10^{13}	17	0.042
$70\,b^3$	17.0	4.8×10^{20}	1.2×10^3	0.16
$100\,b^3$	11.9	2.4×10^{31}	9.3×10^5	1.2
$142\,b^3$	8.4	2.7×10^{46}	1.2×10^{10}	31

5. Discussion

5.1. Three stresses, and the identity of the magnetic phase

Three stresses appear in this paper and must be kept apart. The first is the 147 MPa of Ref. [5]. It is an estimate of the stress at the inclusion surface, obtained from the field through an empirical coefficient; it is not a measured stress, not a resolved shear stress on any slip plane, and not a result of the present calculations. If it were the shear stress on a slip plane it would exceed the 95–105 MPa at which the lower partner of the dislocation pair of Section 3.2 breaks away, so the loaded cells cannot test it by comparison with a threshold; what they test is whether a strained inclusion produces anything of that order in the matrix. The second is what a bonded elastic inclusion strained by 0.194% supplies. The exact solution for a bonded sphere [7]—no sliding at the interface: the atoms on both sides interact through the same potential, and no artificial constraint is applied there—gives 41 MPa of resolved shear inside the particle, falling as $(a/r)^3$ outside (Fig. 4). The third is the field measured in the cell for the same strain held fixed (Section 3.1): 15 MPa of resolved shear just above the crest of the ridge, falling within 80 Å to 2.5 ± 0.5 MPa, the resolution of the calculation; averaged over the width of the cell the peak is 5.0 MPa, because the ridge occupies less than half of that width. The second and the third stress are of one order and together bracket the amplitude entered in Eq. (2). The first exceeds them by a factor of 3.6–10 because $E\varepsilon$ is the stress of a rod held at that strain, not of an inclusion embedded in a matrix that deforms with it.

A separate difficulty concerns the phase itself. In the batch studied, X-ray diffraction identifies the inclusions as $Al_{13}Fe_4$ (written Fe_4Al_{13} in Ref. [5]; the form $Al_{13}Fe_4$ is used throughout here), and the magnetization of the samples shows a hysteresis loop, i.e. the inclusions contain a ferromagnetic constituent [5]; earlier work in the same series gives the composition of the inclusions as Fe_xAl_{1-x} with $x = 86$–87 at.% [3, 6]. Stoichiometric $Al_{13}Fe_4$, however, shows no magnetic order down to 2 K—neither in its magnetization nor in Mössbauer spectra, which probe the magnetic state of the iron atoms directly [19, 20]: the ideal compound does not magnetize and cannot change shape through magnetization (which does not, by itself, guarantee a strictly zero strain of any other origin). The measured ferromagnetism of the inclusions therefore belongs not to the ideal $Al_{13}Fe_4$ lattice but to some other constituent of them—iron-rich regions of the kind noted in Refs. [3, 6], for example—and it is the strain of that constituent under field that matters for the mechanism. That strain has not been measured for the inclusions of this alloy; the 10^{-4} used above is the largest value reported for bulk iron–aluminium alloys [11, 12]. The simulation cell represents the inclusion by $Al_{13}Fe_4$ as the phase identified in this batch, with a known structure [10]; the strain applied to it is taken as a given. Measuring the strain of the actual inclusions under field is the decisive experimental step.

5.2. *Why the mechanism cannot be simulated directly, and what MD can measure instead*

A dislocation behaves as a line under tension. To bow it through a stressed region of size L the stress must exceed about $\mu b/L$, where μ is the shear modulus and b the Burgers vector [18]. If the whole claimed 147 MPa acted as resolved shear stress—the assumption most favourable to the mechanism—this criterion gives $L \approx 52$ nm (about $180\,b$), i.e. a cell of some 2×10^8 atoms once the surrounding matrix is included, against the roughly 10^5 atoms of the cells used here. At 41 MPa the corresponding size is about 0.2 μm, and at 15 MPa about 0.5 μm. The mechanism therefore cannot be simulated directly at the atomic scale. What molecular dynamics can supply is the quantities that a description on the larger scale consumes: the stress at the interface and its decay with distance (Section 3.1), the stresses at which dislocations are set in motion or created (Sections 3.2–3.3), and, through Eq. (2), a check that the whole chain is self-consistent.

5.3. *The memory protocol*

In the experiments the field is absent during creep. The sample is held in the field for 30 min, the field is removed, and only then is the load applied [4, 6]; what is measured is a memory of the exposure. Whatever the field does must therefore persist for tens of minutes after it is gone. An elastic strain of the inclusion cannot: once the field is removed the inclusion returns to its original shape and the stress around it disappears within nanoseconds. The present calculations contain an inclusion held at its strain while the matrix relaxes (the interface cell), one relaxed from it (the free cell) and loaded cells with the inclusion free (unstrained) or held (unstrained and strained); none of them represents the state of the material after the field has been removed. What could last for tens of minutes is a slow rearrangement of atoms—for instance of the Mg and Si atoms that pin the dislocations (Section 3.3)—set in motion while the field acted. Such a rearrangement has not been established here. Supporting it would require the thermal history and the vacancy content of the samples, an estimate of the diffusion rates, and a separate calculation in which the field is switched on, held, switched off and the load then applied. It is therefore stated here as one hypothesis that can be tested, not as a result.

5.4. *Limitations*

Two features of the geometry limit the transfer of these numbers to a real particle. In the cell the inclusion is a ridge—a bar of half-elliptic cross section running the full length of the cell along y (Section 2.1)—rather than a compact particle. The stress outside such a bar falls off with distance more slowly than the $(a/r)^3$ of a compact particle, and its reach—some 30 Å at full strength, 80 Å in all (Section 3.1)—is set by the cross section of the ridge; it is a property of this cell, not of a micron-sized inclusion, whose field would extend correspondingly further. Equation (2) takes its spatial statistics from the three-dimensional solution for a compact particle and its amplitude either from that solution (41 MPa) or from the ridge cell (15 MPa); the two differ by a factor of about 2.7, which is the effect of the geometry and is carried through Table 1 as the difference between its last two columns. A two-dimensional elastic calculation for the ridge alone, with the same strain and elastic constants but in an unbounded matrix of the same stiffness, gives 20 MPa at the ridge surface falling to 5 MPa within 10 Å; the atomistic ridge, held rigid on its rigid layer, gives a smaller surface value and a flatter profile. The atomistic and the continuum description of the ridge thus agree in magnitude, and the difference from the sphere is geometric, not an artefact of the cell.

The interatomic potential [9] was chosen for its coverage of Al, Fe, Mg and Si; how well it reproduces the $Al/Al_{13}Fe_4$ interface, the stress needed to create dislocations, the structure of the dislocation core and the pinning by Mg and Si has not been checked independently here, so the absolute thresholds should be regarded as model-dependent. The residual mismatch between the two lattices is the same in the control and strained cells and cancels to first order in their difference; nonlinear relaxation may prevent exact cancellation.

The thresholds of Section 3.2 are observations in a single cell, with one set of initial atomic velocities and one loading rate; they should be repeated with independent sets of starting velocities and at least a second rate. The pinning result rests on two random arrangements of Mg and Si atoms, only one of which completed the staircase, and one set of starting velocities each; a statistical pinning threshold would require three to five independent arrangements and velocity sets. This is also why the pinning is reported as an observation and no activation volume is extracted from it. The thermostat—the algorithm that holds the cell at 300 K—acts on all mobile atoms and may itself influence the thresholds and the temperature rise when the two dislocations meet and annihilate; the dynamic stage should be repeated either with the thermostat switched off after the temperature has been established, or with it acting only on the atoms at the cell boundaries.

The field is represented only mechanically, as a strain of the inclusion. Mechanisms in which the field acts directly on the dislocation or on the obstacle, as in the theories of Refs. [1, 2], are outside the scope of the calculation. It is worth noting that the changes of barrier height invoked by those theories, 0.01–0.1 eV, correspond through $V^*\Delta\tau$ to 0.97–9.7 MPa at $V^* = 70\,b^3$, and to 0.48–35.9 MPa over the whole range of V^*—overlapping the stresses found here rather than coinciding with them. The stresses themselves are computed from the atomic forces and positions (the virial expression [8]) averaged over slices 4 Å thick parallel to the interface; this is an approximation whose resolution, after the six stress components have been averaged over each slice, is 2.5 ± 0.5 MPa. Finally, Eq. (2) is a separate model relating a local stress to a change of creep rate; it does not remove the dependence of the MD thresholds themselves on the loading rate.

6. Conclusions

The stress that a strained inclusion produces in the matrix was computed in an Al/$Al_{13}Fe_4$ interface cell of 91 428 atoms for an elongation of the inclusion by 0.194% along the field direction at constant volume, the strain corresponding to the earlier estimate of 147 MPa, with the inclusion held at that strain. The resolved shear stress in the matrix reaches 15 MPa just above the inclusion and falls to the far-field level of 2.5 ± 0.5 MPa within 80 Å; averaged over the width of the cell the peak is 5.0 MPa. The analytical solution for a compact particle held at the same strain gives 41 MPa inside it, falling as the inverse cube of the distance, so that for a micron-sized inclusion the same field extends over a fraction of a micron. An inclusion that is not held relaxes most of the strain.

In the loaded cells, a pre-existing pair of dislocations next to the ridge is torn apart at an applied shear of 95–105 MPa with the inclusion free to relax; with the inclusion held, the strained and the unstrained cell coincide within one frame of the ramp, 9 MPa, at every stage; no new dislocation forms at the interface up to the end of the ramp at 400 MPa in the unstrained cell; and a random arrangement of Mg and Si atoms holds a dislocation through 75 MPa. The first picoseconds of every ramp, at zero applied stress, answer the question whether the field of the strained ridge alone moves a dislocation placed next to it: the same partner moves in the same way with the ridge strained and unstrained, driven by the coherency field of the interface, not by the strain of the ridge. These values are specific to the cells, the loading rate and the potential and are not material constants.

The two-scale estimate, Eq. (2), requires a resolved shear stress of 8.4–62.7 MPa at the inclusion surface to raise the creep rate by 25% at $f = 0.00246$ and $V^* = 19\text{–}142\, b^3$. The computed amplitudes, 41 MPa for the particle and 15 MPa for the ridge, reproduce the observed enhancement for V^* of $30\text{–}75\, b^3$, a range that contains the value measured for Al–Mg–Si; 147 MPa acting as resolved shear would exceed the observed enhancement by a factor of at least 2.7×10^3. The elastic mechanism is therefore quantitatively viable if, and only if, the inclusions strain by about 0.2% in the field: the magnetostriction measured for bulk iron–aluminium alloys is twenty times smaller and would give an enhancement below 0.4%.

Because the field is off during the creep test, whatever it does must persist for tens of minutes; an elastic strain cannot, and a slow rearrangement of atoms is indicated as the remaining possibility—a hypothesis not tested by the present calculations. The decisive next steps are the identification of the ferromagnetic constituent of the inclusions, a direct measurement of their strain under field, and a calculation in which the field is switched on, held, switched off and the load then applied.

CRediT authorship contribution statement

I. Mikhailovskiy: Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft. **D. Pshonkin:** Conceptualization, Resources, Supervision, Writing – review & editing.

Data availability

The structure generators, LAMMPS inputs, analysis code and the records behind every number in this paper are in the public repository `https://github.com/iluuua/magnetostriction-plasticity-bounds`. Each figure and table is drawn from a JSON record written by one analysis script: the stress profiles of Fig. 3 and the atomistic curves of Fig. 4 (`stageG10_field_profile.py`), the retained fraction of the imposed strain (`stageG12_eigenstrain_retention.py`), the analytical sphere (`stageG8_eshelby3d.py`) and the two-dimensional solution for the ridge (`stageG17_ridge_continuum.py`), Table 1 (`stageG5_two_scale_bridge.py`), the dislocation trajectories of Fig. 5 and the pinning bound (`stageG6_vstar_analysis.py`, `stageG7_pinning_stats.py`), and the onsets of Section 3.2 (`stageG2_depinning.py`). The two minimised interface cells from which the stress field is measured are included with the code (`data/stageG4_clean`); the renders of Figs. 1 and 2 are produced by `stageG14_render_cells.py`.

Declaration of competing interest

The authors declare no competing financial interests.

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